

Question 1: Will the substrates be provided?

Answer 1: No. The government is purchasing wafers with the specified epitaxial films. Vendors must source epi substrate wafers that meet the specifications indicated for each item.

Question 2: Is wafer size flexible or must be 100 mm?

Answer 2: The delivered wafer size must be 100 mm.

Question 3: Are single side polished (SSP) wafers ok if no double side polished (DSP) substrates are available or the lead time is too long?

Answer 3: The government requires DSP wafers for all substrates to support additional processing within our fab.

Question 4: Pertaining to CLIN 0001, only Si_{0.7}Ge_{0.3} in stack heavily B doped. Si only lightly B doped?

Answer 4: The posted specification is correct. The B dopant concentration is significantly higher in the SiGe layers to improve the selectivity of the later release etch.

Question 5: Pertaining to CLIN 0002, all layers (Si and SiGe_{0.15}, SiGe_{0.3}, SiGe_{0.45}) only lightly B doped?

Answer 5: The posted specification is correct. The B dopant concentration is low for all layers.

Question 6: Pertaining to CLIN 0003, step-function ('staircase' with 4-5. levels) of different dopant concentrations up to a maximum of $5 \times 10^{20} \text{ cm}^{-3}$ or up to at least $5 \times 10^{20} \text{ cm}^{-3}$ (higher concentrations allowed)? Target thickness of entire stack 1micron or less?

Answer 6: This option line item allows the government to purchase "thin" doped Si on Si wafers. The end application for these wafers is reference materials for dopant quantification metrology including secondary ion mass spectrometry (SIMS) and atom probe tomography (APT). These wafers may be specified with up to 5 layers of doped

silicon on the substrate, with no layers being thinner than 10 nm and the entire film stack being less than 1 μm . The following film stack specifications are examples of stacks that are consistent with this requirement:

Example Layer Stack A for CLIN 0003:

Layer	Thickness [nm]	Dopant [$\text{at} \cdot \text{cm}^{-3}$]
3	25 ± 5	Intrinsic
2	50 ± 5	B, 1×10^{16} to 1×10^{17}
1	50 ± 5	B, 1×10^{18} to 1×10^{19}
Substrate		

Example Layer Stack B for CLIN 0003:

Layer	Thickness [nm]	Dopant [$\text{at} \cdot \text{cm}^{-3}$]
2	250 ± 25	B, 1×10^{17} to 1×10^{18}
1	250 ± 25	P, 1×10^{17} to 1×10^{18}
Substrate		

Question 7: Pertaining to CLIN 0003, wafer counts for B and Ph doped wafers?

Answer 7: The lot size is specified when ordered with a minimum order quantity of 10 wafers. Vendors must offer a price for 10 wafers with an additional per wafer cost.

Question 8: Pertaining to CLIN 0004, how many wafers with Boron doped and how many with Phosphorous?

Answer 8: This requirement is for a single dopant layer on the substrate, so the wafer lot would be all one dopant. NIST could elect to execute the option line item twice to get one lot of B doped wafers, and a second lot of Ph doped wafers. In that case, each lot would be subject to the lot size specification in 2.3.

Question 9: Pertaining to CLIN 0004, in terms of the structure, is it exactly the same as 0003, just 10x thicker.

Answer 9: No, CLIN 0004 only allows for a single doped layer whereas 0003 allows film stacks with multiple dopants and concentrations.

Question 10: Pertaining to CLIN 0005, is this a multiple layer structure with various doping level in each layer?

Answer 10: Correct. This option line item allows for layer thickness, isotopics (e.g. natural or ^{28}Si enriched), and dopant to be specified per layer.

Question 11: Pertaining to CLIN 0006, substrate target resistivity?

Answer 11: Substrate resistivity is specified at time of order.

Question 12: Pertaining to CLIN 0006, how many Si/SiGe periods?

Answer 12: This option line item is used to configure film stacks similar to what is already specified in CLIN 0001 and CLIN 0002. The number of layers is numerically limited to 50 (250 nm max film stack thickness divided by 5 nm minimum layer thickness), but this is the upper limit. NIST anticipates film stacks much more similar to CLINs 0001 and 0002.

Question 13: Pertaining to CLIN 0006, is there only one [Ge] in one stack (e.g. 4 different SL flavors)? Or does [Ge] gradually vary between $x=0.1$ to 0.4 in $\text{Si}_{1-x}\text{Ge}_x$ in one stack?

Answer 13: See answer 12. Multiple Ge compositions are allowed within the film stack and they will be specified at time of order on a per layer basis.

Question 14: Pertaining to CLIN 0006, doping in Si and/or SiGe?

Answer 14: Doping is specified on a per layer basis at the time of order.

Question 15: Pertaining to CLIN 0007, Same as line item 0006?

Answer 15: This option line item is for fully relaxed (cubic) SiGe grown on a virtual substrate on Si. These wafers will undergo additional processing to form SiGe on insulator wafers similar to SOIs. Contractors may create the buffer layers of the virtual substrate using their preferred method as long as the thickness of the fully relaxed SiGe layer requirement is met. Similar to the other requirements, dopant, dopant concentration, Ge composition, and SiGe layer thickness are specified at time of order (but anticipated to be $10\text{ }\mu\text{m}$).